

4D to 3D Projection & 4D Rotation

Notes & Mathematical Derivations

1. 4D to 3D Projection

I had options such as:

- Projecting using matrices
- Projecting using the general projection rule to a lesser set of dimension
- Projecting using a hyperplane

Driven by the difficulty and complexity of matrices, and the general projection rule not being as mathematically interesting as the remaining option, I chose **projection using the 3D Hyperplane**.

A 3D Hyperplane, as I have researched, can be represented by the following equation:

$$x + y + z + w = d$$

for $x, y, z, w, d \in \mathbb{R}$.

This is basically a 4D slice in 4D space.

Symbols Used to Explain the Math

P	Point we want to project
C	Camera position
N	Plane normal
Π	Projection plane (Hyperplane)
d	Distance from C to Π
R	Ray
t	Parameter that gets multiplied to vector
P'	Projected point

Mathematical Derivation

The ray R is defined as:

$$R(t) = C + t(P - C)$$

The projection plane Π is defined as:

$$(X - C) \cdot N = \|N\|d$$

for all $X \in \Pi$.

We now need t :

$$(C + t(P - C) - C) \cdot N = \|N\|d$$

$$t = \frac{\|N\|d}{(P - C) \cdot N}$$

Substituting t back into the ray equation gives the projected point P' :

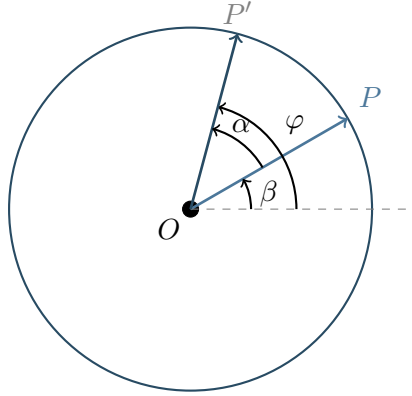
$$P' = C + \frac{\|N\|d}{(P - C) \cdot N} \cdot (P - C)$$

2. Rotation in 4D

There are 4 dimensions, and each rotational plane changes two, so there are 6 different rotational planes:

$$\mathbf{XY}, \mathbf{XZ}, \mathbf{XW}, \mathbf{YW}, \mathbf{ZW}, \mathbf{YZ}$$

To rotate a point on a rotational plane, we need to first find the origin of the plane. Conventionally, you do $(P - O)$, with P being the point and O being the origin.



Using $\beta = \text{atan2}()$, we calculate the current angle, and using:

$$d = \sqrt{(P - O) \cdot (P - O)}$$

we calculate distance.

α is the angle we want to add:

$$\varphi = \alpha + \beta$$

The rotated coordinate position:

$$P' = (\cos(\varphi) \cdot d, \sin(\varphi) \cdot d)$$